Module 03

VI Editor & Bash Scripting

DevOps end to end services & solutions

```
#!/bin/bash
vi file.txt
```

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vi file.txt
```

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Agenda

- VI Editor Introduction
- VI Modes & Shortcuts
- Linux Shell Fundamentals
- Bash Scripting: Variables, Inputs, Math
- Bash Logic: If/Else & Loops
- Hands-on Lab & Solution

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What is VI / VIM?

The Power Editor

VI (Visual) is a standard text editor available on almost every Linux system.

- **Universal:** Works on all distros, even via SSH.
- **Efficient:** Keyboard-driven, no mouse needed.
- **VIM:** (Vi IMproved) is the modern, feature-rich version.

VI (Visual Editor)

Original Unix editor, minimalist, modal, no syntax highlighting, single undo.

Released 1976.

VIM (Vi IMproved)

Enhanced version, rich features, syntax highlighting, multiple undo/redo, plugins, graphical mode (gVim).

Modern standard.

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VI Modes

Command Mode

The default mode. Keys perform commands (move, copy, delete).

Press ESC to return here.

Insert Mode

Allows you to type and edit text like a normal editor.

Press i to enter.

Last Line Mode

Save, quit, or search.

Press : from Command Mode.

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Essential Shortcuts

Navigation & Edit

- **i** : Insert before cursor
- **dd**: Delete (cut) current line
- **yy**: Yank (copy) current line
- **p** : Paste below cursor
- **u** : Undo

Save & Quit

- **:w** : Write (save) file
- **:q** : Quit
- **:wq** : Write and Quit
- **:q!** : Quit without saving (force)

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Bash Scripting

Automating the Shell

A shell script is a file containing a series of commands.

- **Shebang:** #!/bin/bash tells the system which interpreter to use.
- **Permissions:** Must be executable (chmod +x script.sh).
- **Execution:** Run with ./script.sh

```
#!/bin/bash
# My first script
echo "Hello World"
mkdir new_folder
cd new_folder
touch file.txt
```


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Variables & Inputs

Handling Data

- **Variables:** Define with NAME="Value" (no spaces). Use with $NAME.
- **Input:** Use read to get user input.
- **Math:** Use $((...)) or expr.

```
#!/bin/bash
NAME="DevOps"
echo "Welcome $NAME"
echo "Enter your age:"
read age
next_year=$((age + 1))
echo "Next year you will be $next_year"
```


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Questions?

Thank you for participating.

Feel free to reach out:
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